

Confounding factors and biases abound when predicting molecular biomarkers from histological images

Received: 30 August 2024

Accepted: 16 January 2026

Published online: 02 March 2026

 Check for updates

Muhammad Dawood^{1,2,3}✉, Kim Branson⁴, Sabine Tejpar⁵,
Nasir Rajpoot^{2,6} & Fayyaz ul Amir Afsar Minhas^{1,2,7}✉

Deep learning models that infer clinically relevant biomarker status from tissue images are being explored as rapid and low-cost alternatives to molecular testing. Here we show, through statistical analysis across multiple cancer types, datasets and modelling approaches, that the datasets used to train these models contain strong dependencies between biomarkers and clinicopathological features, which prevent models from isolating the effect of a single biomarker and lead them to learn confounded signals. Consequently, their prediction accuracy varies substantially with the status of codependent biomarkers and clinicopathological variables, and for several biomarkers, the gain over what a pathologist can already infer from routine histopathological features, such as grade, remains modest. These findings indicate that current approaches are not yet suitable as substitutes for molecular testing but can support triage or complementary decision-making with caution. Unconfounded biomarker prediction will require models that learn causal rather than correlational relationships between biomarkers and tissue morphology.

Fuelled by developments in computational pathology, several studies have proposed methods to predict clinically relevant biomarkers^{1–9}, such as gene mutations and expression levels, directly from routine haematoxylin and eosin (H&E)-stained whole-slide images (WSIs)^{1–4,10}. These approaches take a WSI as input and predict the status of clinically relevant biomarkers such as microsatellite instability (MSI), hormonal receptors or mutations in *TP53*, *BRAF*, *KRAS*, *EGFR* and other genes, as their target. Such methods are typically motivated by two main objectives: first, to identify or mine histological patterns associated with specific biomarkers¹¹, and second, to rule out certain biomarkers from routine WSIs, avoiding the need for additional stains or molecular testing, which can be tissue-destructive, costly and associated with longer turnaround time¹². For example, the accurate prediction of

MSI^{1,4,13} and mutations in genes such as *BRAF*¹ and *KRAS/NRAS*¹⁰ from WSIs can inform personalized treatment decisions while reducing cost and waiting time compared with sequencing¹².

Several methods have demonstrated that, in specific cancers^{10,14,15}, biomarker status^{2,3,16} and alterations in certain genes are predictable from WSIs using deep learning pipelines trained in a weakly supervised fashion on imaging and molecular data from The Cancer Genome Atlas (TCGA) or other similar data repositories, such as the Clinical Proteomic Tumour Analysis Consortium (CPTAC)¹⁷. However, for most biomarkers, the prediction accuracy of these methods remains low, with the area under the receiver operating characteristic curve (AUROC) values ranging from 0.50 to 0.90. Moreover, the true generalization of such methods to external datasets is further challenged by factors such

¹Predictive Systems in Biomedicine Lab, Department of Computer Science, University of Warwick, Coventry, UK. ²Tissue Image Analytics Centre, University of Warwick, Coventry, UK. ³Nuffield Division of Clinical Laboratory Sciences, Radcliffe Department of Medicine, University of Oxford, Oxford, UK. ⁴Artificial Intelligence and Machine Learning, GSK, San Francisco, CA, USA. ⁵Digestive Oncology, Department of Oncology, Katholieke Universiteit Leuven, Leuven, Belgium. ⁶Histofy Ltd, Coventry, UK. ⁷Warwick Cancer Research Centre, University of Warwick, Coventry, UK.

✉e-mail: imuhdawood@gmail.com; Fayyaz.Minhas@warwick.ac.uk

as mutation prevalence, limited multicentric data, class imbalance between positive (mutated or high expression) and negative (wild-type or low expression) cases, quality of WSIs (such as pen markings and tissue tears) and domain shifts¹⁸. In this Article, we demonstrate that even if these challenges have been handled, there are underlying fundamental issues that require addressing.

In a WSI, disease phenotypes manifest as different visual patterns arising from the interaction of multiple codependent genes rather than from a single factor. These interactions are often characterized by patterns of mutual exclusivity or co-occurrence among molecular factors^{19–21}. A detailed discussion of the pathobiological origins of these interactions is provided in Section 1.1 of the Supplementary Information. Despite this, current approaches primarily focus on predicting the status of individual biomarker or gene mutation from WSIs, neglecting codependencies between covariates. Although several recent studies^{5,22} have used multi-output models and leveraged representations from multimodal foundation models to predict biomarker status from WSIs; these studies remain limited to optimizing aggregated accuracies and do not extend to assessing the stability of model performance across patient groups stratified by the status of a codependent biomarker.

In this study, we show that overlooking interdependencies among biomarkers can influence the predictive performance of machine learning (ML) models. We argue that interdependencies among biomarker statuses in the training data, when ignored during model development, can lead to models capturing the aggregated influence of multiple interdependent biomarkers rather than patterns linked to a single biomarker. Moreover, this could also spuriously inflate or deflate models' apparent performance in certain subgroups when the interdependency structure among molecular factors shifts in the test cohorts. Finally, when clinicopathological variables (for example, tumour mutational burden (TMB) or tumour grade) are themselves associated with biomarker status, models may rely on phenotypes associated with these correlated variables as predictive proxies, instead of capturing the intended biological signal.

To illustrate these effects, we first analysed interdependencies among biomarkers by assessing their patterns of mutual exclusivity and co-occurrence²³. We then use permutation testing and stratification analysis to demonstrate failure modes of WSI-based predictors by showing that their accuracy for a given biomarker varies substantially when conditioned on the status of other biomarkers. We also highlight the need for appropriate causal adjustments in WSI-based predictors to ensure reliable inferences necessary for informing clinical decisions, such as treatment selection and pathobiological understanding. To this end, we propose a stratification-based evaluation framework to report bias and support the development of more transparent and trustworthy ML models to advance WSI-based precision diagnostics.

Results

Data and study design

We analysed the limitations of existing ML approaches for predicting molecular biomarkers (for example, mutations, genomic instability indicators and protein expression) from H&E stained WSIs. A high-level concept diagram of these approaches is provided in Fig. 1. We hypothesize that interdependencies among biomarker statuses and clinicopathological variables in the training data, and the disregard of such associations during model development, bias ML models towards relying on aggregated influences of multiple factors in WSIs rather than patterns linked to individual biomarkers. To illustrate this, we retrospectively analysed $n = 8,221$ patients with breast cancer (BRCA), colorectal cancer (CRC), endometrial cancer (UCEC) and lung cancer across four cohorts for which WSIs and/or molecular information (for example, receptor status, gene mutations and so on) were available (Methods). These include: TCGA ($n = 2,683$), Molecular Taxonomy of Breast Cancer International Consortium (METABRIC; $n = 2,433$)^{24,25}, Memorial Sloan Kettering Cancer Centre (MSK; $n = 2,486$)^{26,27} and DFCI

($n = 619$)²⁸. Using these datasets, we performed the four major steps listed below:

- (1) An analysis of the interdependency among biomarkers and somatic mutation status of genes in samples;
- (2) Training deep learning models to predict biomarker status from WSIs;
- (3) Stratification analysis and permutation testing to assess whether the model trained to predict a certain biomarker is biased by the status of other biomarkers or clinicopathological variables;
- (4) An analysis of the added value of using ML models in predicting various biomarkers over and above the pathologist-assigned grade.

Drawing from established methods in gene functional analysis^{20,21,29–31}, we quantified the interdependency among molecular factor labels across patients by evaluating their pattern of co-occurrence and mutual exclusivity. We used log odds ratios (LOR) to quantify these relationships, where positive LOR values indicate co-occurrence, and negative values indicate mutual exclusivity. Statistical significance was assessed with a two-sided Fisher's exact test, and the resulting P values were corrected for multiple hypothesis testing.

To assess whether biomarker interdependencies introduce bias into WSI-based models, we analysed three deep learning algorithms with different principles of operation: attention-based (CLAM³²), graph neural network-based (SlideGraph³³) and a WSI-level multimodal foundation model (TITAN²²). These algorithms represent existing ML approaches that do not explicitly consider interdependencies between prediction variables. As CLAM and SlideGraph³³ rely on a patch-level encoder, we trained them with two different encoders: CTransPath³⁴ (trained on histology images) and ShuffleNet³⁵ (trained on ImageNet)³⁶ to minimize encoder-specific bias. For each biomarker, we train these models with both encoders on the TCGA cohort using fourfold cross-validation and report AUROC as a performance metric. We further evaluated the trained models on two independent validation cohorts, CPTAC³⁷ and the Australian Breast Cancer Tissue Bank (ABCTB)³⁸. Finally, we used WSI-level representation from a multimodal foundation model (TITAN)²², trained on 330,000 image–text pairs, under the hypothesis that these embeddings better capture biomarker-related morphology, and trained both single-output and multi-output biomarker predictors on them.

To investigate whether WSI-based biomarker prediction models are confounded by the interdependency among molecular factors or clinicopathological variables (for example, histological grade or TMB), we performed a stratification analysis and permutation testing. For each model, we define two types of variable: the prediction variable, which is the biomarker the model is trained to predict, and stratification variables, which are biomarkers or clinicopathological features showing significant mutual exclusivity or co-occurrence with the prediction variable and may act as confounders (identified in step 1). The motivation for considering interdependent variables as confounders is that they may be associated with a shared phenotypic pattern in WSIs, which the model can exploit as proxies for the prediction variable, potentially leading to biased predictions when such signals are absent or decoupled at test time. To detect such confounders, we evaluate model performance at two levels: (1) across the entire cohort and (2) within subgroups defined by stratification variables. Examining model performance within these subgroups allows us to isolate the effect of the prediction variable from confounders. If the model truly captures prediction variable specific patterns in WSIs, its subgroup-level performance should closely match the cohort-level performance. By contrast, substantial differences between subgroups and overall performance indicate the influence of confounding effects or Simpson's paradox^{39,40}. To quantify these effects, we perform permutation testing and report their statistical significance.

For example, to evaluate whether the performance of a WSI-based predictor for oestrogen receptor (ER) status (prediction variable) is influenced by *TP53* mutation status (stratification variable),

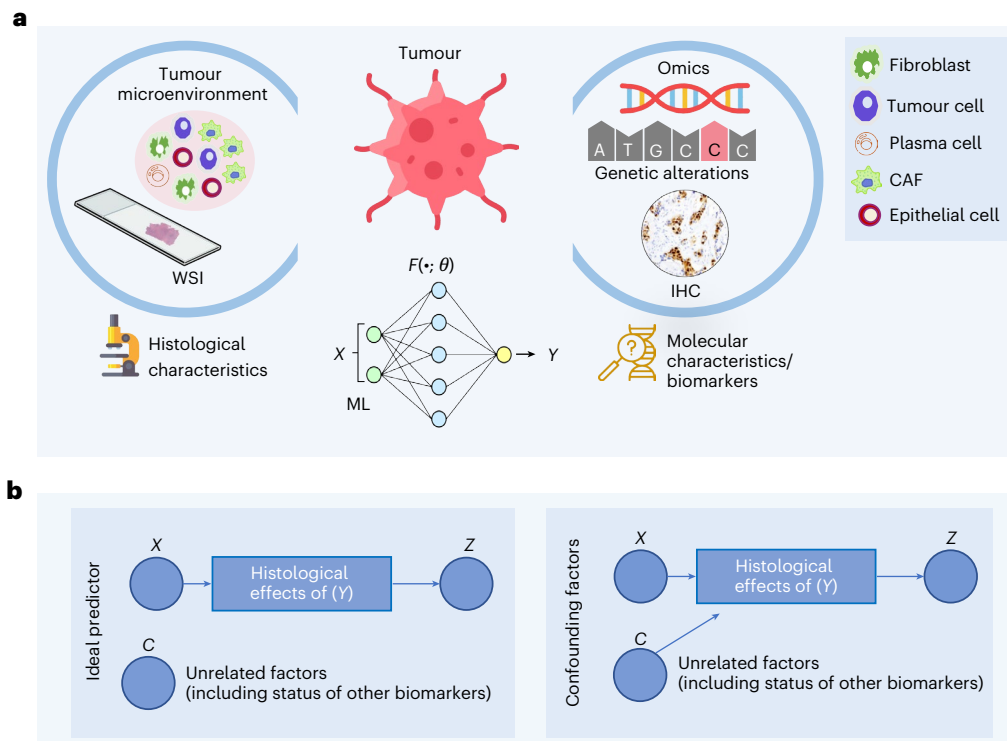


Fig. 1 | Conceptual framework of ML methods that infer molecular biomarker status from histology WSIs. a. The ML-based prediction of molecular biomarkers from WSIs involves using training data of WSIs with known biomarker statuses. The ML model accepts the representation of a WSI (X) as input and predicts the status of a certain biomarker (Y) as the target. **b.** An ideal predictor should be able to predict the status of a molecular biomarker from histological effects of that biomarker contained in the WSI, and its output (Z) should be

independent of unrelated confounding factors (lumped into a variable C) as shown in the simplified causal diagram. Conversely, if the predictor's output is dependent not only on the histological effects of (Y) but also on other confounding factors (for example, histological grade, TMB or status of other biomarkers), then the prediction is confounded because the model is relying on these additional covariates rather than solely on the effects of (Y). Credit: icons in **a**, [Flaticon.com](https://www.flaticon.com).

we first divide the cohort into two subgroups on the basis of the stratification variable: patients with a *TP53*-mutant status and patients with a *TP53* wild-type status. We then compute the AUROC of the ER predictor within each of these subgroups. Finally, we compare these subgroup-level AUROCs to the model's overall AUROC across the entire cohort. A substantial difference between subgroup and cohort-level AUROCs indicates a potential bias, suggesting the model captures the combined effects of ER and *TP53* rather than ER-specific features alone. To establish statistical significance, we run a permutation test with 10,000 trials (see Methods for more details). This definition of the 'prediction variable' (ER status in this example) and the 'stratification variable' (*TP53* status in this example) will be used consistently in subsequent results and figures to ensure clarity. Repeating this across alternative stratification variables (for example, grade and TMB) provides a systematic way of detecting the influence of confounding factors on different WSI-based models.

To assess the added value of ML models in predicting various biomarkers over and above pathologist-assigned grades, we used a support vector machine with one-hot encoded histological grades to predict various clinical biomarkers following the same protocols used for weakly supervised models.

Biomarker statuses show significant interdependencies and variations

Our analysis revealed significant interdependencies ($P \ll 0.05$) among biomarkers across cancer types (Fig. 2 and Supplementary Fig. 1). In BRCA, elevated ER and progesterone receptor (PR) expression co-occur with mutations in *CDH1*, *MAP3K1* and *PIK3CA*, but not with *TP53*, which is mutually exclusive with *CDH1*, *GATA3*, *MAP3K1* and *PIK3CA*⁴¹. In CRC, MSI-high (MSI-H) cases frequently carry *BRAF*, *ATM*, *ARID1A* and *RNF43*

mutations and are less likely to harbour *KRAS* mutations; *BRAF*-mutant tumours also show higher TMB and show co-occurrence with *ATM*, *RNF43* and *ARID1A*. Similar patterns of interdependencies are also observed in UCEC and lung adenocarcinoma (LUAD) (Supplementary Fig. 1). For instance, in UCEC, *PTEN* mutations co-occur with *APC*, *ATM*, *JAK1*, *KRAS* and *ARID1A*, whereas in LUAD, *STK11* mutations co-occur with *KEAP1* but rarely with *EGFR*.

Our analysis further showed that, within the same tissue type, biomarker associations can vary across datasets, showing sampling variations. In the TCGA-BRCA cohort, *MAP3K1* mutations showed mutual exclusivity with *AKT1* and *ARID1A*, whereas in the METABRIC cohort, they showed a tendency towards co-occurrence (Fig. 2). ER status and high TMB showed mild co-occurrence in the TCGA-BRCA cohort but mutual exclusivity in the METABRIC cohort. In the TCGA-CRC cohort, *BRAF*-mutant tumours were significantly less likely to harbour *TP53* mutations, whereas this association is less pronounced in the DFCI cohort and lacks statistical significance. Similar cross-dataset differences were observed in UCEC and LUAD (Supplementary Fig. 1). For instance, in TCGA-LUAD, *BRAF* and *STK11* showed a weak tendency towards mutual exclusivity, whereas in the MSK cohort, they showed a weak tendency towards co-occurrence.

These results show that biomarker statuses are significantly interdependent and that their association patterns can vary across datasets. Consequently, ML models trained on WSIs may learn composite phenotypes driven by multiple interdependent biomarkers, introducing cohort-specific biases and limiting their generalizability to unseen cases.

Prediction of biomarkers and gene alterations from WSIs

To demonstrate that the ML models analysed in the study were properly trained, we report biomarker prediction performance across

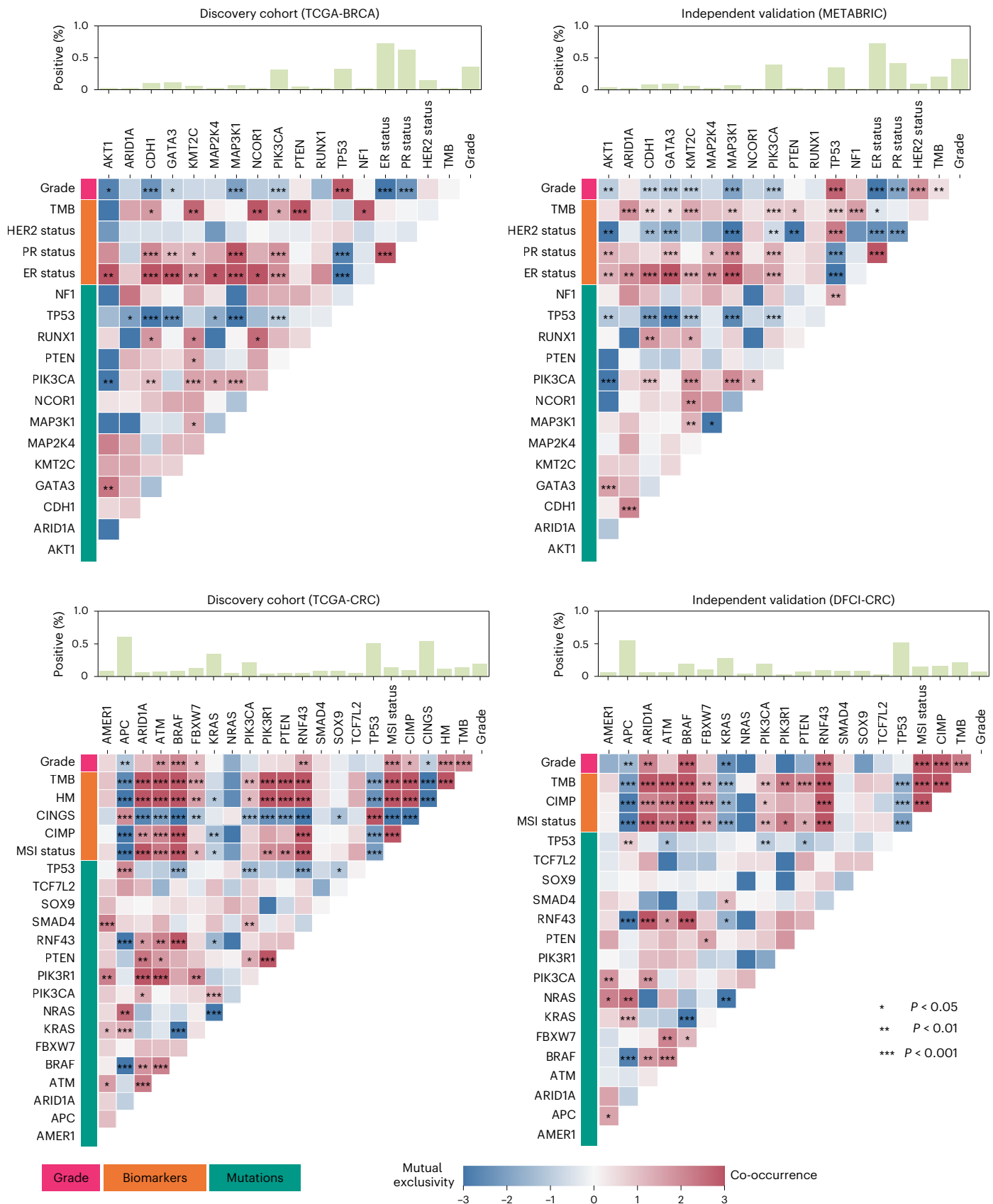


Fig. 2 | Heat maps showing associations of biomarkers and gene mutation statuses across tissue types and datasets. The heat maps display a set of biomarkers and genes along the axes, with cell colours within the heat map showing the strength of association (dark red colours for co-occurrence and dark blue for mutual exclusivity). Cells marked with asterisks indicate statistically significant associations (Benjamini–Hochberg FDR-corrected P values from

two-sided Fisher's exact test $P < 0.05$). The top bar above each heat map shows the percentage of cases mutated for a specific gene in case of gene mutations, whereas for biomarkers, it indicates the percentage of patients with elevated ER, PR and HER2 in case of breast tumours, high MSI, hypermutation and CIMP activity and CIN for colorectal tumours. CING5, chromosomally instable versus genome stable; HM, hypermutated.

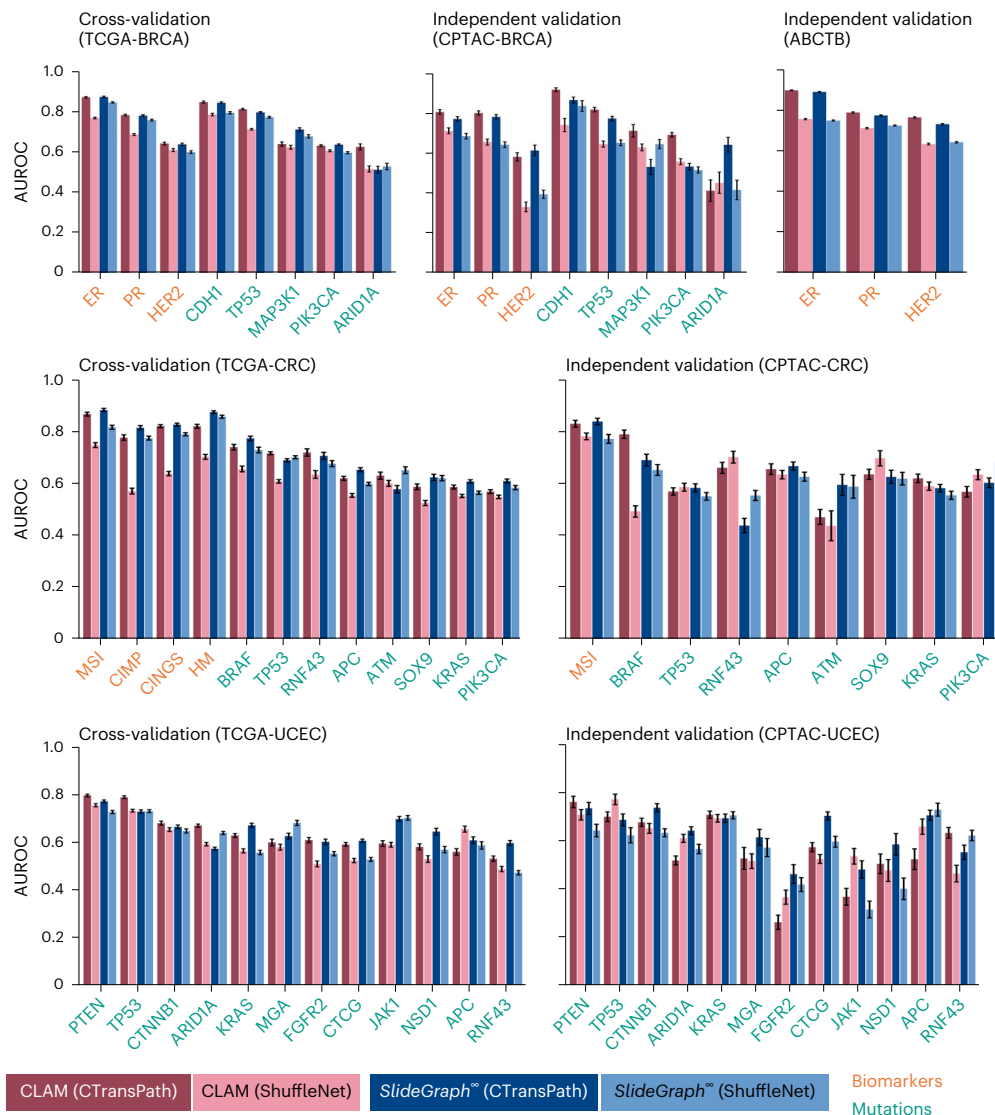


Fig. 3 | Quantitative results of weakly supervised models in predicting biomarkers/mutations from WSIs across different cancer types. The plots show the AUROC for two weakly supervised models (CLAM and SlideGraph[∞]), each trained with two different patch-level encoders: ShuffleNet, a convolutional neural network-based encoder pretrained on natural images, and CTransPath, a transformer-based model pretrained on WSIs through self-supervised learning. For each biomarker or gene mutation, the comparative predictive performance

for these four model-encoder combinations is shown. Dark and light red bars correspond to CLAM with CTransPath and ShuffleNet, respectively, whereas dark and light blue bars correspond to SlideGraph[∞] with CTransPath and ShuffleNet, respectively. Bar heights represent mean AUROC values, whereas error bars indicate the 95% confidence (two-sided, using Student's *t*-distribution) calculated across 100 class-stratified bootstrap sampling runs. Bar labels are colour-coded, with yellow denoting biomarkers and green denoting mutations.

algorithms, feature embeddings and modelling approaches (Fig. 3 and Supplementary Figs. 1 and 2). Different model configurations achieved AUROCs >0.80 for multiple biomarkers in both cross-validation and independent validation cohorts.

In BRCA, CLAM with CTransPath features predicts receptor status with average AUROCs of 0.87 and 0.90 for ER and 0.79 and 0.78 for PR, in cross-validation (TCGA-BRCA) and independent validation (ABCTB) cohorts, respectively. Similar AUROCs were observed for SlideGraph[∞] (CTransPath). These models also inferred gene mutations with high accuracy; for example, CLAM (CTransPath) predicted *CDH1* and *TP53* mutations with AUROCs of 0.88 and 0.82 in TCGA-BRCA and 0.91 and 0.82 in CPTAC-BRCA, respectively.

Beyond breast tumours, these models also achieved high AUROC values for predicting biomarkers and gene mutations in CRC, lung cancer and UCEC (Fig. 3 and Supplementary Fig. 2). For instance, SlideGraph[∞] (CTransPath) predicted MSI status in CRC with an AUROC of 0.89 in TCGA-CRC (cross-validation) and 0.84 in CPTAC-CRC (independent validation). A strong predictive performance was also

observed for other biomarkers, including *BRAF*, CpG island methylator phenotype pathway (CIMP), CING5 and hypermutation status (Fig. 3).

Apart from weakly supervised approaches, single-output and multi-output models trained on TITAN WSI-level feature representation showed roughly similar performance (Supplementary Fig. 3). For example, the multi-output model predicts the ER and PR status of TCGA-BRCA cases with an AUROC of 0.89 and 0.81, respectively, closely matching the AUROC values of models trained under the single-output setting (ER 0.89 and PR 0.79).

These results confirm the proper training of these models. Next, on the basis of AUROC, we selected the best model for each biomarker and assessed the influence of biomarker interdependencies through permutation testing and stratification analysis.

Interdependence in biomarker status leads to entangled histology phenotypes captured from WSIs

Our confounding factor analysis reveals that WSI-based predictors are strongly influenced by biomarker interdependencies. Across multiple

biomarkers, the higher cohort-level AUROCs achieved by these models drop substantially in subgroups defined by the statuses of various stratification variables (Fig. 4 and Supplementary Figs. 4–7). For example, SlideGraph[∞] predicts colorectal tumours' MSI status (the 'prediction variable') with an AUROC of 0.88 (0.873–0.886). However, when the same patient set is divided into hypermutated and non-hypermutated subgroups (the 'stratification variable'), the AUROC for MSI status prediction drops to 0.72 within each subgroup. A similar effect is observed in stratification by other biomarkers showing co-occurrence with MSI (for example, CIMP activity, hypermutation and *APC* statuses) and those showing mutual exclusivity (for example, *BRAF* and *CINGS*) (Fig. 4).

These observations extend beyond colorectal tumours and are evident in biomarker predictors of breast and endometrial tumours, irrespective of the specific model architecture, feature embeddings or training methodology used. For instance, in breast tumours, the performance of the ER predictor substantially declines in cases with *GATA3*, *CDH1* and *PIK3CA* mutations (Fig. 4). Likewise, the ER predictor's AUROC drops substantially in both PR-positive and negative cases, as well as in *TP53*-mutant and wild-type cases. Similar trends are apparent for PR, *TP53*, *CDH1* and *PIK3CA* predictors (Fig. 4). This trend of inconsistent subgroup performance is also observed for other single- and multi-output models, such as those utilizing TITAN WSI-level feature representation (Supplementary Figs. 5–7). For example, the AUROC of the ER predictor drops from 0.89 to 0.57 in single-output settings, whereas it drops from 0.88 to 0.58 under multi-output settings.

These results suggest that the biomarker prediction from ML models is contingent on the status of other interdependent biomarkers, and these models are probably relying on composite phenotypes arising from potentially interacting biomarkers rather than learning biomarker-specific morphology.

WSI-based biomarker prediction is confounded by histology grade

WSI-based models predict breast tumour receptor status (ER, PR) with high cohort-level AUROCs of 0.87 and 0.79 in the TCGA-BRCA cohort, and 0.90 and 0.78 in the ABCTB cohort, respectively. However, the stratification analysis by tumour grade reveals marked subgroup-level performance drops (Fig. 5). The ER predictor AUROC drops to 0.76 for medium-grade cases in both cohorts, and the PR predictor AUROCs in low and medium-grade cases drop to 0.59 and 0.69 in the TCGA-BRCA cohort and to 0.65 and 0.73 in the ABCTB cohort. Mutation predictors show similar grade-specific performance declines; for example, AUROC of the *TP53* predictor drops from 0.81 (cohort-level) to 0.73, 0.73 and 0.72 for low-, medium- and high-grade cases. These patterns extend beyond breast tumours and are evident in the mutation predictors of endometrial tumours, irrespective of model architecture, feature embeddings or training methodology (Fig. 5 and Supplementary Fig. 8). For example, *TP53* predictors trained on TITAN WSI-level embeddings also show performance drops in high-grade cases, with AUROCs decreasing from 0.83 to 0.77 in single-output settings and from 0.86 to 0.77 in multi-output settings.

Our analysis further shows that the apparent AUROCs of WSI-based models are sensitive to shifts in biomarker-grade associations between training and test cohorts. For example, in high-grade UCEC cases, the *TP53* predictor attains an AUROC of 0.70 in the TCGA cohort but only 0.36 in the CPTAC cohort, a pattern consistent with a shift in *TP53*-grade relationship from strong co-occurrence in the training cohort to moderate mutual exclusivity in the test cohort. Similarly, in low-grade cases, the ER predictor achieves an AUROC of 0.96 in the ABCTB cohort compared with a cross-validation AUROC of 0.90 in TCGA-BRCA, probably reflecting a stronger ER-grade association in ABCTB than in TCGA. Consistent with these, single- and multi-output models trained on TITAN WSI-level feature representations showed similar sensitivity (Supplementary Fig. 8). For example, in TCGA-UCEC, *TP53* AUROC

drops from 0.83 to 0.77 in high-grade cases for the single-output model and from 0.86 to 0.77 for the multi-output model. In CPTAC-UCEC, where the grade–mutation association differs, the drop in AUROC is more pronounced, from 0.61 to 0.53 for the single-output model and from 0.74 to 0.60 for the multi-output model.

The confounding influence of grade is further supported by experiments in which, for selected biomarkers, we trained separate models for grade 1, 2 and 3 patients; these grade-specific models attained lower AUROCs than the pooled model (Supplementary Table 1). For example, in TCGA-BRCA, the *TP53*-grade-specific predictors achieved AUROCs of ~0.73 compared with 0.84 for the pooled model, and ER and PR showed similar reductions. To evaluate whether these disparities could be attributed to demographic differences, we examined the demographic balance between biomarker-positive and biomarker-negative cases and found moderate racial differences (Supplementary Table 2). We therefore repeated the grade-stratified experiment only on patients in a single racial subgroup (white). The same trends persisted (Supplementary Table 3); for example, the ER predictor trained only on grade 1 cases achieved an AUROC of 0.66, substantially lower than the pooled AUROC of 0.85, suggesting that demographic factors are unlikely to drive these performance differences (Supplementary Table 3).

These results, reminiscent of Simpson's paradox, indicate that WSI-based biomarker prediction models rely heavily on grade-associated morphology rather than biomarker-specific phenotypic signatures, making them less generalizable to external cohorts where grade–biomarker associations differ from those in the training data.

The added predictive power of biomarker predictors beyond pathologist grade assignments

Our analysis shows that the status of several biomarkers across cancer types can be inferred with accuracy higher than expected from pathologist-assigned grade, and in several cases, approaches the performance of deep learning models. In BRCA, grade-based ER and PR classifiers achieved AUROCs of 0.76 and 0.70 in the TCGA-BRCA cohort and 0.79 and 0.71 in the ABCTB cohort, respectively (Fig. 6). Grade also predicts *TP53* mutations with an AUROC of 0.75, nearly matching the 0.81 achieved by weakly supervised ML models. Similar AUROC patterns were seen for *TP53* and *PTEN* predictors in the TCGA-UCEC and CPTAC-UCEC cohorts. These results suggest that, for some biomarkers, ML algorithms offer limited additional predictive value over pathologist-assigned grade (Fig. 3). The strong grade–biomarker association also risks ML models linking grade-associated phenotypic differences to biomarker status; therefore, WSI-based models are expected to exceed this grade-derived baseline and establish robust phenotype–genotype associations that are independent of tumour grade.

WSI-based biomarker prediction is confounded by the density of mutations in other genes

WSI-based models infer *BRAF* and *TP53* mutations in colorectal tumours (TCGA-CRC) from WSIs with high confidence, achieving AUROCs 0.774 (0.764–0.785) and 0.717 (0.711–0.722), respectively (Fig. 7a). However, stratification analysis reveals a significant challenge: for cases with low mutation density in genes other than *BRAF* (denoted as $TMB_{\overline{BRAF}}$), the *BRAF* predictor accuracy drops to an AUROC of 0.65 (Fig. 7a). Similarly, the *TP53* predictor AUROC drops to 0.50 for high TMB cases. In the CPTAC-CRC cohort, similar trends were observed, with *BRAF* and *TP53* predictors' performance dropping in low and high TMB cases, respectively. In addition, *APC* and *KRAS* mutation predictors are also influenced by TMB. This observation also extends to UCEC, where the *PTEN* predictor achieved AUROCs of 0.803 in TCGA-UCEC and 0.731 in CPTAC-UCEC but drops to 0.63 and 0.32 for low TMB cases in the respective cohorts (Fig. 7a).

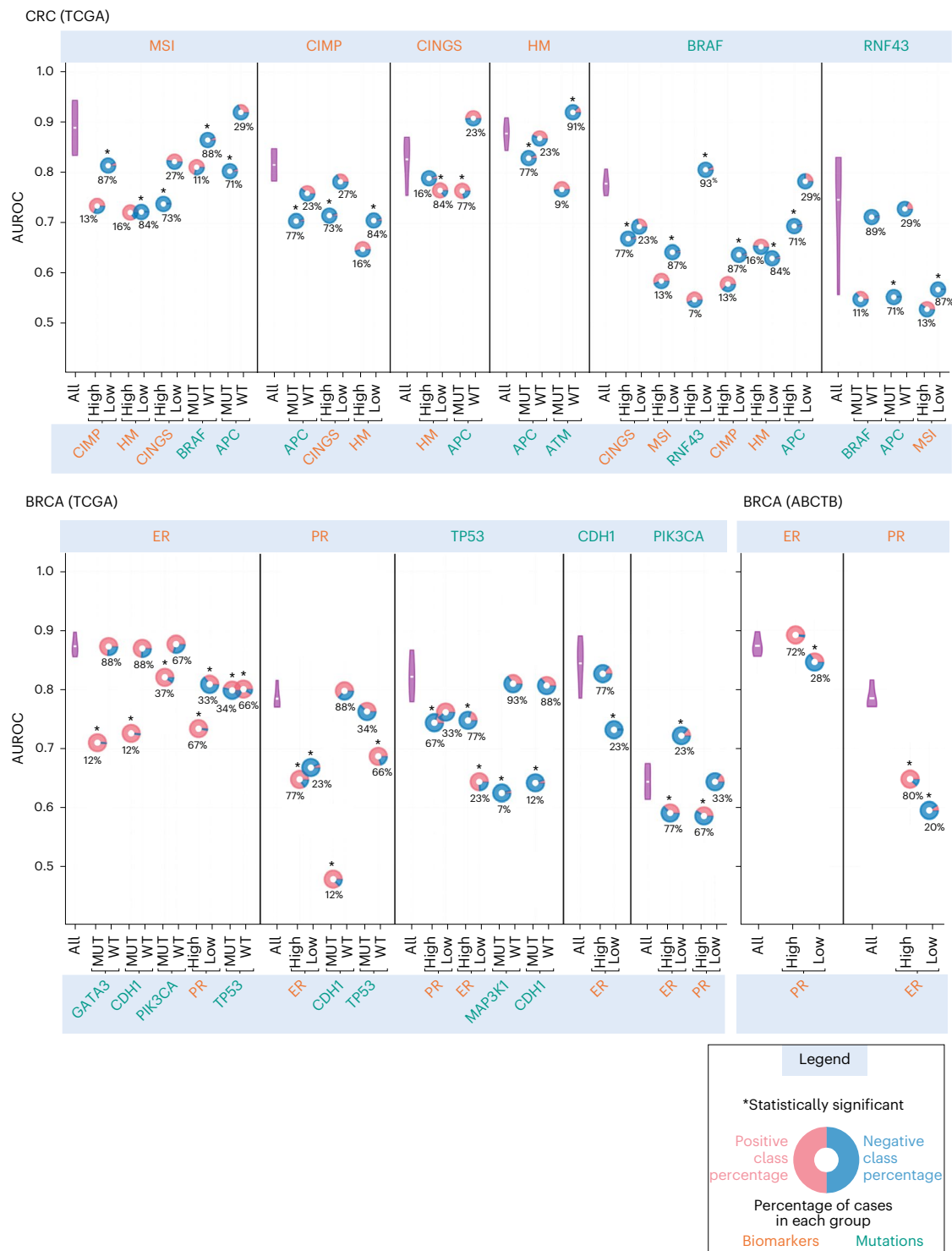


Fig. 4 | Plots showcasing stratification analysis of several WSI-based biomarker predictors concerning other interdependent biomarkers.

AUROC values are illustrated on the y-axis, with the top x-axis indicating the prediction variables and the bottom x-axis showing the stratification variables. The predictive performance of each predictor on all the cases in the cohort (denoted by 'All' in the plot) over 100 bootstrap runs is shown using a violin plot, whereas its performance in different stratification groups is depicted with a doughnut chart, with the centre representing the AUROC values. The horizontal white line inside each violin marks the mean of the distribution.

Doughnuts marked with an asterisk at the top indicate statistically significant variation in results in the stratification analysis (Benjamini–Hochberg FDR-corrected $P \ll 0.05$). The percentage values at the bottom of the doughnut indicate the proportion of positive (MUT/high) or negative (WT/low) cases relative to the status of the stratification variables. Red and blue colours in each doughnut indicate the proportion of positive and negative cases in each stratified group concerning prediction variables. MUT, mutated; WT, wild-type.

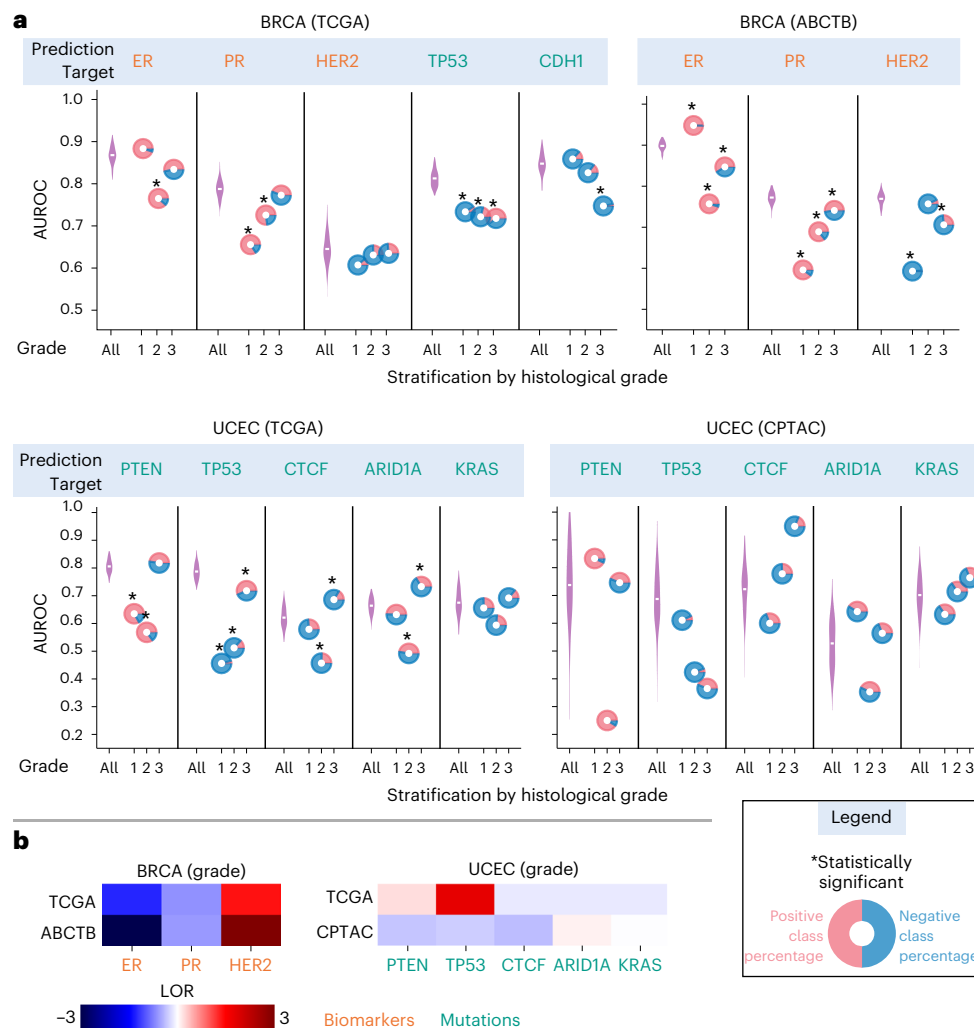


Fig. 5 | Plots illustrating the biased predictive performance of WSI-based biomarker predictors across patients with different histological grades through stratification analysis. a, In the plots, AUROC values are illustrated on the y axis, with the top x axis indicating the prediction variables and the bottom x axis showing the patient stratification with respect to histological grade. The predictive performance of each predictor on all the cases in the cohort (denoted by 'All' in the plot) over 100 bootstrap runs is shown using a violin plot, whereas its performance in a group of patients with a certain histological grade is depicted with a doughnut chart, with the centre representing the AUROC values. The horizontal white line inside each violin marks the mean of the distribution.

Doughnuts marked with an asterisk at the top indicate statistically significant differences in results (Benjamini–Hochberg FDR-corrected P values from two-sided permutation testing $P \ll 0.05$). Red and blue colours in each doughnut indicate the proportion of positive and negative cases in each stratified group in relation to prediction variables. **b**, Heat maps highlighting the shift in the association structure between histological grade and biomarker status across two distinct datasets. The colour intensity reflects the strength of association, with dark red indicating strong co-occurrence and dark blue indicating strong mutual exclusivity.

We further show that varying associations between TMB and biomarker status across datasets significantly influence the prediction accuracy of WSI-based predictors. In CRC, the association between *KRAS* mutation and TMB is slightly stronger in the CPTAC-CRC cohort compared with the TCGA-CRC cohort (Fig. 7b). This stronger association could explain the *KRAS* predictor's significantly improved prediction accuracy (AUROC: 0.83) in high TMB cases in the CPTAC-UCEC cohort, compared with an AUROC of 0.63 for high TMB cases in the TCGA-CRC cohort. This analysis suggests that the model's predictions are not only influenced by the *KRAS* mutation status, which is the target prediction variable, but also by the overall TMB, which affects the prediction accuracy.

Discussion

Deep learning models trained on routine WSIs of H&E-stained tissue sections are increasingly discussed as rapid and cost-effective tools to infer molecular biomarker status in patients with cancer. In this

study, we identified key limitations of these approaches for clinical and preclinical use, in particular, their failure to account for biomarker interdependencies during model training and inference. Through statistical analysis, we first demonstrated significant interdependencies among molecular factors across tissue types and datasets (TCGA, METABRIC, MSK and DFCI), manifested as patterns of mutual exclusivity and co-occurrence that reflect both pathobiological and spurious associations. Subsequently, using permutation testing and stratification analysis, we showed that these associations in the training data lead to models whose predictions for a given biomarker are contingent on the status of other codependent biomarkers. For example, the PR predictor showed a marked drop in performance in *CDH1*-mutant cases, with AUROC decreasing from 0.79 to 0.50. This decline in subgroup performance suggests that the current ML models cannot fully disentangle biomarker-specific signals from the multifaceted influence of molecular characteristics and other factors on tissue phenotypes in WSIs.

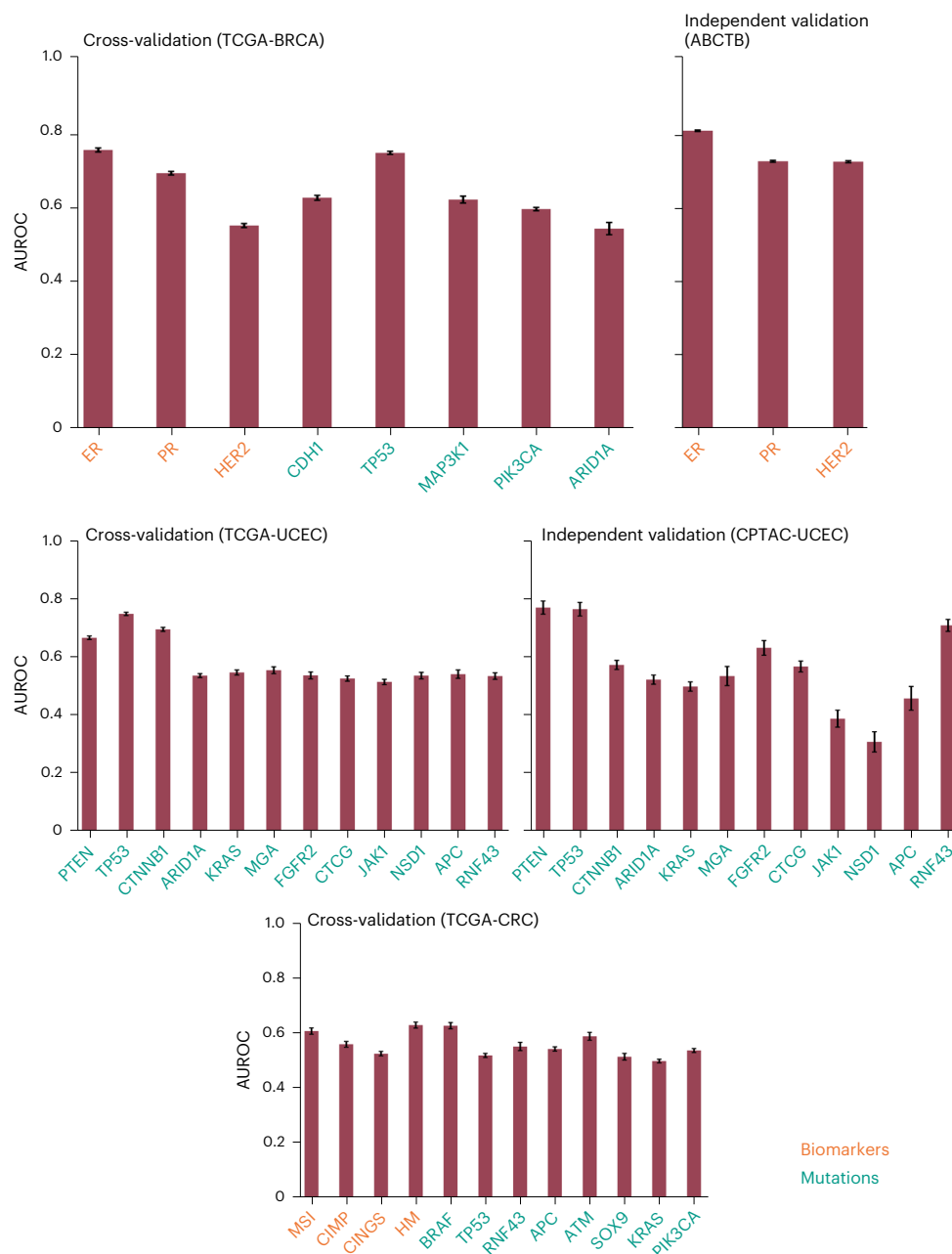


Fig. 6 | Quantitative results of prediction of biomarkers/mutations using pathologists' assigned grade. The plots illustrate the AUROC achieved by a support vector machine classifier trained to predict a biomarker/gene mutation from one-hot encoded histological grades. Bar heights represent mean

AUROC values, whereas error bars indicate the 95% confidence (two-sided, using Student's *t*-distribution) calculated across 100 class-stratified bootstrap sampling runs. Bar labels are colour-coded, with yellow denoting biomarkers and green denoting mutations.

The inability of WSI-based models to discern biomarker-specific signals has direct clinical implications when codependent biomarkers have divergent therapeutic roles. An example is the *BRAF*-MSI association in CRC. Our analysis shows that MSI predictions from WSI-based models are contingent on *BRAF* status, with AUROCs dropping in both *BRAF*-mutant and wild-type subgroups, and a similar pattern was observed for the *BRAF* predictor when stratified by MSI status (Figs. 2 and 4 and Supplementary Fig. 7). This reflects their well-known biological co-occurrence: MSI-H CRCs frequently harbour *BRAF*V600E mutations, whereas MSI-stable CRCs rarely harbour *BRAF* mutations. Crucially, however, MSI-H and *BRAF* mutations have distinct therapeutic implications. MSI-H is a strong predictor of the response to immune checkpoint inhibitors such as pembrolizumab or nivolumab, whereas *BRAF*V600E mutations are targeted using *BRAF* and MEK inhibitors in

combination with *EGFR* blockade. Combinations of immunotherapy and *BRAF* inhibitors are currently being tested for the double mutant. A model that cannot disentangle MSI-H from *BRAF* status may achieve high aggregate AUROC but lacks clinical utility, as confusing the two would misguide treatment selection. This example underscores the broader need for bias-aware evaluation: predictors must be assessed not only for overall accuracy but also for their ability to distinguish correlated biomarkers with divergent therapeutic pathways⁴².

Beyond the influence of biomarker interdependencies, we showed that these models exploit prominent grade- or TMB-associated features in WSIs as proxies for biomarker prediction (Figs. 5 and 7). In breast tumours, AUROCs of ER and *TP53* predictors drop markedly within grade-stratified subgroups and shifts in the grade-biomarker association across cohorts lead to apparent

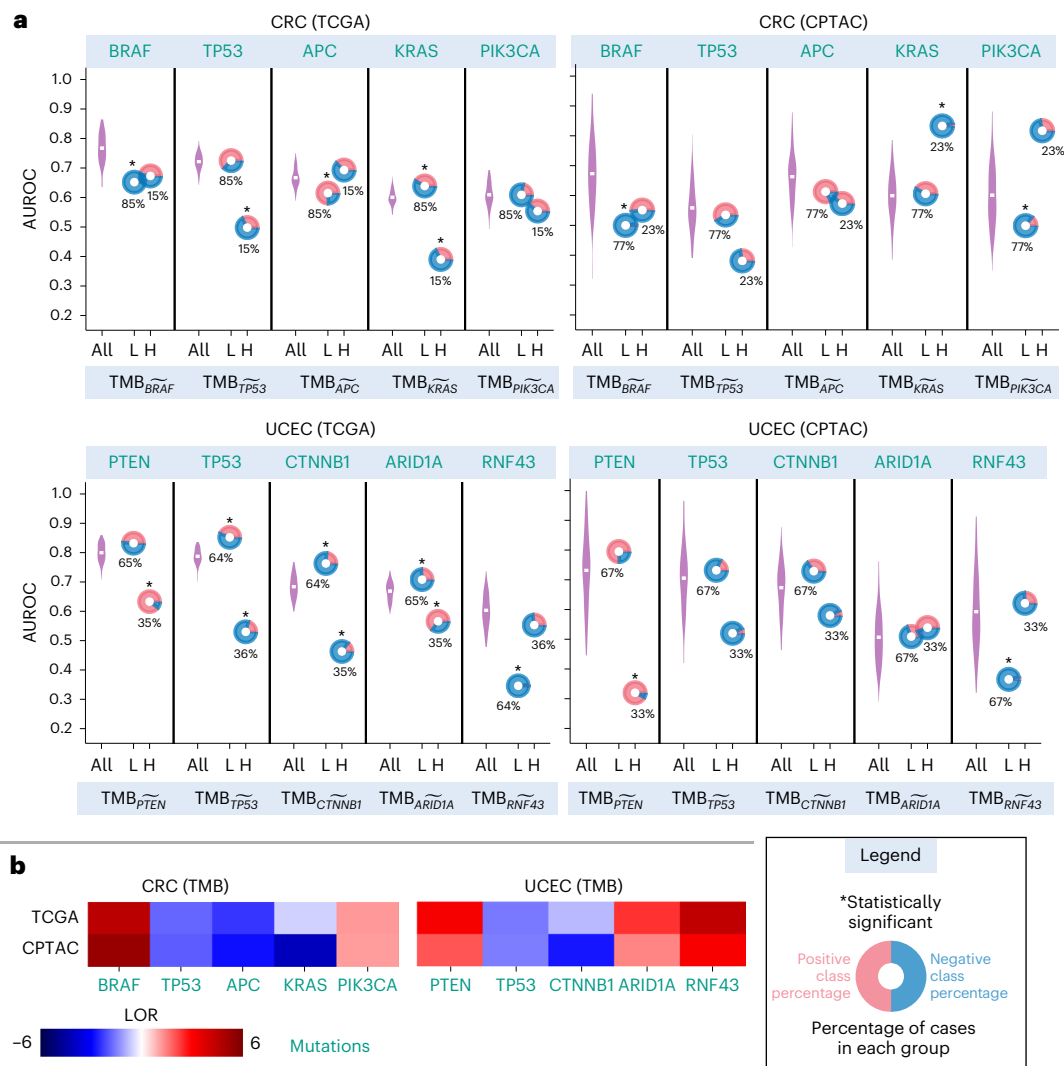


Fig. 7 | Plots illustrating the biased predictive performance of WSI-based biomarker predictors across patients with different TMBs through stratification analysis. a, AUROC values are plotted on the y axis, with the top x axis indicating the prediction variables and the bottom x axis showing patients' stratification with respect to TMB. The predictive performance of each predictor on all the cases in the cohort (denoted by 'All' in the plot) over 100 bootstrap runs is shown using a violin plot, whereas its performance in patients with high and low TMB is depicted with a doughnut chart, with the centre representing the AUROC values. The horizontal white line inside each violin marks the mean of the

distribution. Doughnuts marked with an asterisk at the top indicate statistically significant variation in results (Benjamini–Hochberg FDR-corrected P values from two-sided permutation testing $P \ll 0.05$). Red and blue colours in each doughnut indicate the proportion of positive and negative cases in each stratified group based on prediction variables. **b,** Heat maps highlighting the shift in the association structure between TMB and gene mutations across two distinct datasets. The colour intensity reflects the strength of association, with dark red indicating strong co-occurrence and dark blue indicating strong mutual exclusivity.

improvements or declines in accuracy. Likewise, TMB-stratified analysis shows substantial AUROC declines for *BRAF*, *TP53* and other markers, with shifts in TMB–biomarker association across cohorts influencing apparent accuracy. These patterns, observed across different ML models and feature representations, reflect a broader challenge in computational pathology: models tend to exploit confounding variables (grade, TMB) and conflate them with biomarkers of interest (for example, ER, PR, *TP53* and *PTEN* status), thereby obscuring true genotype–phenotype relationships, limiting generalizability and introducing bias. This also raises concerns about their suitability for routine clinical use, because substantial heterogeneity in biomarker profiles can exist among tumours with the same grade or TMB, and both grade and TMB can evolve over the disease course or treatment. Consequently, models that rely on these prominent features are vulnerable to distribution shifts and may produce inconsistent predictions for the same patient at different time points, irrespective of the true biomarker status.

These findings underscore the need to interpret external validation results with caution. In our analysis, the ER predictor achieved a high AUROC of 0.87 in cross-validation on TCGA-BRCA and 0.90 in a larger independent cohort (ABCTB), which could be interpreted as an excellent generalizability of the model. However, upon closer examination, we found that the apparent improvement in AUROC was largely driven by a stronger grade-ER association in the ABCTB than in the training cohort. Moreover, within grade-stratified subgroups, the predictive performance of this sophisticated ER predictor was not substantially more informative than a simple grade-based classifier. This illustrates that external validation must be complemented by bias-aware evaluations, such as grade- and TMB-stratified analyses, before claiming clinical utility.

The confounding influence of biomarker interdependencies and clinicopathological variables (for example, grade and TMB) on current WSI-based biomarker prediction suggests that current models are not yet ready to replace genomic testing in routine care. Instead, they are

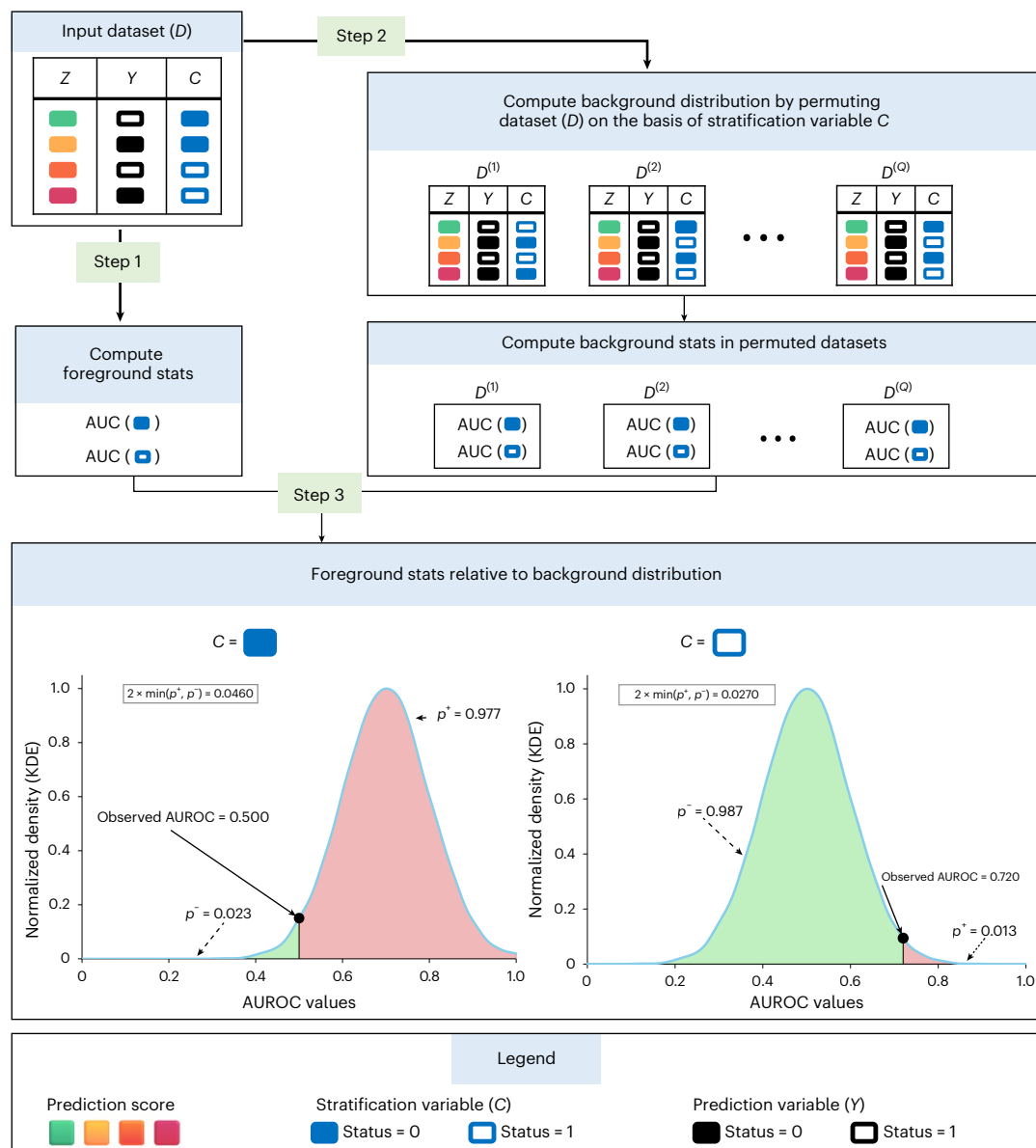


Fig. 8 | Conceptual diagram showing the workflow of permutation testing and stratification analysis to assess bias in WSI-based models. The algorithm takes as input a dataset containing prediction scores (Z), ground truth labels (Y) and a confounding or stratification variable (C). In step 1, the algorithm computes foreground statistics, such as AUROC within each stratum defined by the values of C. In step 2, the algorithm permutes C multiple times (represented by Q), generating permuted datasets $D^{(1)}$, $D^{(2)}$, ..., $D^{(Q)}$. AUROCs are computed in each

permuted dataset, where any association C and Y has been randomized to form a null distribution reflecting expected model performance under the assumption of no association between C and Y. In step 3, the algorithm compares the observed AUROCs against null distributions to assess how extreme they are. If they lie in the tails, the effect of C is considered statistically significant, and a two-sided multiple hypothesis corrected P value is computed. KDE, kernel density estimation.

better positioned for triaging, screening or supplementary decision support, provided that their performance is rigorously assessed and key clinical decisions remain supported by confirmatory testing. To ensure true clinical utility, we suggest bias-aware evaluation, including reporting grade- and TMB-stratified metrics and subgroup calibration rather than relying solely on aggregate AUROC (Figs. 4, 5 and 7 and Fig. 8). Our findings also have implications for studies and trials that link disease phenotypes to biomarkers or assess treatment response conditioned on biomarker status. In both contexts, establishing robust relationships requires that the biomarker of interest is not tightly coupled to cohort-specific covariates, as such dependencies can lead to false conclusions. To mitigate this risk, we recommend: (1) preserving variation in the target biomarker relative to correlated variables during enrolment; (2) prespecifying stratification factors (for example,

grade, TMB, site, key comutations) and conducting prospective subgroup analyses; (3) including a dependency-aware analysis plan (for example, stratified permutation tests, subgroup confidence intervals, comparison with simple clinical baselines such as grade-only models); and (4) conducting per stratum power calculations rather than only aggregate targets.

Although ML methods for predicting biomarker status from WSI have limitations, they can still provide substantial value. They can facilitate research and hypothesis generation by uncovering associations between histology and molecular factors, particularly in tissue-limited or retrospective scenarios where running additional assays is not feasible. WSI-based models also offer a scalable and cost-effective surrogate for large-scale preclinical and translational studies and can serve as rapid prescreening tools in early phase trials or resource-constrained

settings⁴³. In drug development, they can help narrow the pool of candidates for more resource-intensive molecular analyses and, with appropriate safeguards and clinician oversight, can support triage by guiding decisions on when confirmatory testing is essential⁴³. To support safe use, we recommend bias-aware evaluation and interpretation of prediction results, including subgroup-stratified metrics and permutation-based checks, and comparisons against simple baselines such as grade-based classifiers.

Although predicting biomarker status from routine H&E WSIs may appear to be a simple image-to-label mapping, it is considerably more complex because phenotypes in WSIs are rarely driven by a single factor and instead reflect combined effects of multiple codependent molecular factors. Our analyses show that current approaches, including single and multi-output models, as well as ML and graph-based methods across different feature representations, fail to reliably learn biomarker-specific genotype–phenotype mapping; instead, they exploit aggregated phenotypes of interdependent biomarkers or cohort-specific association as proxies for prediction. This results in biased models whose performance drops across patients' strata defined by codependent variables. These findings motivate the need for methods that formalize the problem as causal, structured multi-label learning: explicitly encode dependencies among biomarkers in the label space, learn disentangled image representations guided by conditional-independence objectives, mitigate confounding via causal adjustment and counterfactual data augmentation and optimize for invariance and distributional robustness, coupled with evaluation protocols based on conditional metrics and subgroup calibration^{44,45}.

Although we demonstrated the generalizability of our findings across multiple cancer types, datasets and modelling approaches, this study still has limitations. First, our analyses were limited to H&E WSIs with WSI-level (coarse) labels, and we did not evaluate immunohistochemistry (IHC) slides or models trained with fine-grained labels (for example, spatial omics supervision). Second, although we used a large multicentre dataset ($n = 8,221$), prospective studies are needed to define clinical and deployment guidelines. Third, we note that learning disentangled genotype–phenotype mapping using ML will probably require combinatorially richer datasets with the exhaustive coverage of comutation or biomarker-pair combinations than current cohorts; however, curating such datasets would necessitate significant long-term efforts. Last, we suggested several methodological directions for ML researchers to explore and words of caution for clinicians, but their effectiveness remains to be established; it is premature to recommend definitive clinical guidelines.

Methods

Ethics statement

All samples used in the study were obtained with research consent and ethics approvals as indicated in the consent and ethics statements for TCGA, METABRIC, COAD-DFCI, MSK-LUAD, CPTAC and ABCTB.

Patient cohorts

We analysed data of four cancer types (BRCA, CRC, LUAD and UCEC), sourced from six cohorts: TCGA⁴⁶, METABRIC^{24,25}, COAD-DFCI²⁸, MSK-LUAD²⁶, CPTAC and ABCTB. Biomarkers and gene mutation status information, except for the ABCTB cohort, were collected from cBioportal³⁰. WSIs of formalin-fixed paraffin-embedded (FFPE) H&E-stained tissue for TCGA atlas cases were collected from TCGA^{46,47}, whereas for CPTAC atlas cases, they were retrieved from The Cancer Imaging Archive (TCIA)¹⁷. Within the ABCTB cohort, WSIs and receptor status (ER, PR and human epidermal growth factor receptor 2 (HER2) status) information were available for 2,303 patients. In terms of biomarkers, for breast tumours, ER, PR and HER2 status were recorded. For colorectal cases, MSI, hypermutation (HM), chromosomal instability (CIN) and CIMP activity statuses were documented. TMB information was available for all cases across all cancer types and cohorts.

Intergene mutational dependency analysis

We analysed the interdependency between the mutational status of genes using the LOR. Given the status of two biomarkers A and B, in a given dataset, we calculated the LOR as follows:

$$\text{LOR} = \log_2 \left(\frac{((n_A + n_B)(n_{\sim A} + n_{\sim B}))}{((n_A + n_{\sim B})(n_{\sim A} + n_B))} \right).$$

In the above equation, n_A and n_B denote the number of cases that are positive for A and B, respectively, whereas $n_{\sim A}$ and $n_{\sim B}$ denote the number of cases that are negative for those biomarkers. A higher positive LOR between gene pairs indicates mutation co-occurrence (that is, if one gene is mutated, the other is likely to be mutated), whereas a negative value signifies mutual exclusivity of mutation (that is, if one gene is mutated, the other is less likely to be mutated).

In addition to the LOR analysis, we statistically assessed the inter-dependence among the mutational status of different genes using a two-sided Fisher's exact test. All gene pairs were enumerated, and a Fisher's exact test was performed on each pair. Subsequently, we reported the multi-hypothesis corrected P values for each pair using the Benjamini–Hochberg method, with a significance threshold set at $P \ll 0.05$.

Prediction of biomarkers and gene alteration from WSI

We assessed the predictability of biomarkers and gene alteration status from WSIs within their respective cohorts using two algorithms with different principles of operation: CLAM³² and SlideGraph³³. To avoid drawing conclusions specific to a certain approach or type of features, the predictive performance of both algorithms was evaluated over different types of feature: deep features (a convolutional neural network-based encoder trained on ImageNet)³⁶ and self-supervised features (a transformer-based model trained on histology images in a self-supervised manner)³⁴. Our predictive pipeline comprises three main steps: (1) preprocessing of WSIs, (2) embedding of WSI patches, (3) biomarkers and gene mutation prediction from WSIs using CLAM and SlideGraph³³.

Preprocessing. In our preprocessing pipeline, utilizing a U-Net-based segmentation model from TIAToolbox⁴⁸, we first segment viable tissue areas of each WSI and exclude regions with artefacts (pen-marking, tissue folding and so on). The model-generated tissue mask highlights informative tissue areas within the WSI using a pixel value of 1, whereas those with a value of 0 represent background or regions with artefacts. Leveraging these tissue masks, from each WSI, we extract patches of size 512 pixels $\times$ 512 pixels and 1,024 pixels $\times$ 1,024 pixels at a spatial resolution of 0.50 microns-per-pixel. We selectively keep patches (both benign and tumour) that have more than 40% viable tissue in terms of pixel proportion.

Feature representation. We utilized various encoders to extract feature representation from WSI patches. Specifically, we used ShuffNet³⁵ pretrained on ImageNet³⁶ as a patch-level encoder to extract the 1,024-dimensional feature representation (deep features) from WSI patches of size 512 pixels $\times$ 512 pixels. Moreover, we also extracted a 768-dimensional self-supervised feature representation from each patch of size 1,024 pixels $\times$ 1,024 pixels using CTransPath (a transformer-based self-supervised model trained on histology images)³⁴.

Predictive models. We trained SlideGraph³³ and CLAM for predicting the status of different clinical biomarkers using both deep features and self-supervised features. In case of SlideGraph³³, we first construct a graph representation of the WSI and then pass the WSI graph to a graph neural network for predicting the status of a certain biomarker as output. In cases where patients had multiple WSIs, we constructed a serial graph incorporating all WSIs and predicted the target label accordingly. In the case of CLAM, we bag all the WSIs

belonging to the same patient and then predict the target label from the WSI bag.

Apart from these weakly supervised models, we also analysed alternative modelling approaches using feature representations from TITAN²², a state-of-the-art multimodal foundation model trained on more than 330,000 WSIs paired with pathology reports. We leveraged TITAN-derived features to train both single-output and multi-output models for biomarker prediction. In the single-output settings, WSI-level features were fed into a logistic regression model to predict the status of a single biomarker. In the multi-output settings, we used a multilayer perceptron (MLP) model that takes WSI-level representations as input and simultaneously predicts the status of all biomarkers as output. The model architecture consists of a single hidden layer that projects the input to half its dimension, followed by a rectified linear unit activation function and then an output layer. The model was trained using a pairwise ranking loss function³³.

Training and validation of image-based predictors

We trained and evaluated the performance of both SlideGraph[∞] and CLAM using fourfold cross-validation, in which the dataset is partitioned into four 75/25 non-overlapping splits. In each cross-validation run, the model is trained on 75% data, and the performance of the trained model is then assessed on the 25% test set. From the training dataset, we randomly select 10% of the data for validation. We trained the model for 300 epochs on the training set, with a batch size of 8 and a learning rate of 0.001 using the adaptive momentum-based optimizer⁴⁹. To limit overfitting, we stop the model training if its performance on the validation cohort is not improving over ten consecutive epochs. We quantitatively assess model performance on the test set using AUROC as a performance metric. Our motivation for using AUROC as the primary metric was twofold: (1) it allows us to maintain comparability with existing literature and align with established benchmarking practices, and (2) it serves as a threshold-free, rank-based statistic for bias detection, enabling subgroup evaluation and stratified permutation testing. We used the same train, validation and test splits for both SlideGraph[∞] and CLAM.

Baseline predictors based on histology grade

To assess the predictability of biomarkers and gene mutation status on the basis of histology grade, we used a linear model (specifically, a support vector machine). This model uses the one-hot encoded histological grade as input to predict the status of a certain clinical biomarker as the target. We followed the same training and evaluation protocols used for our weakly supervised models. We quantify the model's predictive performance using AUROC as a performance metric.

Stratification analysis and permutation testing for evaluating confounding effects

To investigate whether WSI-based biomarker prediction models are confounded by biomarker interdependency or clinicopathological variables (for example, histology grade or TMB), we used a stratification-based permutation testing approach. A high-level conceptual overview of the approach is shown in Fig. 8, and complete algorithmic details are presented in Supplementary Table 4. Using the procedure outlined in that table, we evaluate the robustness of model performance to confounding influence from biomarkers or clinicopathological features that exhibit mutual exclusivity or co-occurrence with the prediction variable (hereafter referred to as stratification variables).

Let $D = \{(Z_i, Y_i, C_i) \mid i = 1, \dots, N\}$ denote the dataset for a given test cohort, where $Z_i \in \mathbb{R}$ is the score generated by a WSI-based model trained to infer the prediction variable $Y_i \in \{0, 1\}$. The variable $C_i \in V_C$ denotes the stratification variable (for example, status of a codependent biomarker or clinicopathological feature), and V_C is the set of all unique values that C can take (for example, mutant or wild-type for mutation status). The objective is to assess whether the model's

performance in predicting Y is conditionally independent of the stratification variable C .

For each subgroup $v \in V_C$, we compute a stratified performance measure using AUROC as a performance metric. We define the foreground metric as $M_{C=v} = \text{AUROC}(\{(Z_i, Y_i) \mid C_i = v\})$ which reflects model performance restricted to a subgroup where $C = v$.

To determine whether $M_{C=v}$ significantly deviates from what would be expected under the null hypothesis, that is, when the model predictions Z are independent of C , we conduct a stratified permutation test. Let $Q = 10,000$ be the number of permutations. For each permutation trial $q = 1, \dots, Q$, we define a permutation function $\pi_q: \{1, \dots, N\} \rightarrow \{1, \dots, N\}$, which randomly shuffles the assignment of C while preserving the correspondence between Z and Y . A permuted dataset is constructed as: $D^{(q)} = \{(Z_i, Y_i, C_{\pi_q(i)}) \mid i = 1, \dots, N\}$ and for each $v \in V_C$, we compute the permuted AUROC: $M_{C=v}^{(q)} = \text{AUROC}(\{(Z_i, Y_i) \mid C_{\pi_q(i)} = v\})$. The collection $\{M_{C=v}^{(q)}\}_{q=1}^Q$ forms the empirical null distribution of AUROC values under the assumption of no dependence between the prediction variable Z and stratification variable C .

To quantify whether the observed stratified performance $M_{C=v}$ is significantly different from the null distribution, we compute a two-sided P value:

$$p_v^+ = \frac{1}{Q} \sum_{q=1}^Q I(M_{C=v}^{(q)} \geq M_{C=v}),$$

$$p_v^- = \frac{1}{Q} \sum_{q=1}^Q I(M_{C=v}^{(q)} \leq M_{C=v}),$$

$$p_v = 2 \times \min(p_v^+, p_v^-).$$

In the above equations $I(\cdot)$ is the indicator function. The term p_v^+ captures the upper-tail P value (the proportion of permuted AUROCs greater than or equal to the observed value), and p_v^- captures the lower-tail P value (the proportion of permuted AUROCs less than or equal to the observed value). The final P value p_v quantifies the statistical evidence that the model's predictive performance in the subgroup $C = v$ differs from what would be expected under the null hypothesis (no association between the model predictions and the stratification variable). A lower value of p_v suggests that the model's predictions are influenced by the stratification variable, implying reliance on proxy features rather than those directly linked with the prediction variable^{50,51}.

Using the stratified permutation test discussed above, we examined three key factors that could introduce bias into an ML model: first, the bias due to interdependency among biomarkers and the somatic mutation status of genes in the training dataset; second, a likely bias due to patients' tumour histological grades; and third, an expected bias due to the TMB of a patient with cancer. To assess the influence of interdependence among biomarker statuses on model predictive performance, we select the model with the highest AUROC score for each biomarker and run a permutation test, treating other biomarkers with codependent statuses as confounding variables. Subsequently, to analyse the influence of histological grade on WSI-based biomarker predictors, we use a similar approach, utilizing histology grade as a confounding variable. Finally, to evaluate the impact of TMB on histology image-based biomarker predictors, we first calculate patient-level TMB excluding genetic alterations of the gene of interest used for prediction, then use this TMB_{voi} as a confounding variable. On the basis of TMB_{voi} , we divide the patients into low and high TMB cases using a threshold of ten mutations per megabase.

As this procedure is repeated across multiple stratification variables and subgroups, all P values p_v are corrected for multiple hypothesis testing using the Benjamini–Hochberg procedure. Adjusted P values below a false discovery rate (FDR) threshold of 0.05 are considered statistically significant.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

WSIs of TCGA patients used in the study can be downloaded from the NIH Genomic Data Commons Portal at this link: <https://portal.gdc.cancer.gov/>. The genomic data and clinical data of patients in TCGA, METABRIC, COAD-DFCI, MSK-LUAD and CPTAC cohorts can be downloaded from cBioPortal at <https://www.cbioportal.org/>. ABCTB data and images were obtained from the ABCTB. The corresponding author can be contacted to facilitate access to ABCTB data.

Code availability

All the experiments were conducted using Python using PyTorch Geometric library and TIAToolbox. Code and documentation of all Python scripts used in the study are available via GitHub at <https://github.com/imuhdawood/HistBiases>. Any additional information required to reproduce the data reported in this work is available from the corresponding author upon request.

References

- Bilal, M. et al. Development and validation of a weakly supervised deep learning framework to predict the status of molecular pathways and key mutations in colorectal cancer from routine histology images: a retrospective study. *Lancet Digit. Health* **3**, e763–e772 (2021).
- Lu, W. et al. SlideGraph+: whole slide image level graphs to predict HER2 status in breast cancer. *Med. Image Anal.* **80**, 102486 (2022).
- Wagner, S. J. et al. Transformer-based biomarker prediction from colorectal cancer histology: a large-scale multicentric study. *Cancer Cell* **41**, 1650–1661 (2023).
- Kather, J. N. et al. Deep learning can predict microsatellite instability directly from histology in gastrointestinal cancer. *Nat. Med.* **25**, 1054–1056 (2019).
- McCaw, Z. et al. Machine learning enabled prediction of digital biomarkers from whole slide histopathology image. Preprint at medRxiv <https://doi.org/10.1101/2024.01.06.24300926> (2024).
- Zhao, Y. et al. Deep learning using histological images for gene mutation prediction in lung cancer: a multicentre retrospective study. *Lancet Oncol.* **26**, 136–146 (2025).
- Campanella, G. et al. Real-world deployment of a fine-tuned pathology foundation model for lung cancer biomarker detection. *Nat. Med.* **31**, 3002–3010 (2025).
- Echle, A. et al. Deep learning in cancer pathology: a new generation of clinical biomarkers. *Br. J. Cancer* **124**, 686–696 (2021).
- Coudray, N. et al. Classification and mutation prediction from non-small cell lung cancer histopathology images using deep learning. *Nat. Med.* **24**, 1559–1567 (2018).
- Saldanha, O. L. et al. Self-supervised attention-based deep learning for pan-cancer mutation prediction from histopathology. *NPJ Precis. Oncol.* **7**, 1–5 (2023).
- Fu, Y. et al. Pan-cancer computational histopathology reveals mutations, tumor composition and prognosis. *Nat. Cancer* **1**, 800–810 (2020).
- Lim, C. et al. Biomarker testing and time to treatment decision in patients with advanced nonsmall-cell lung cancer†. *Ann. Oncol.* **26**, 1415–1421 (2015).
- Echle, A. et al. Clinical-grade detection of microsatellite instability in colorectal tumors by deep learning. *Gastroenterology* **159**, 1406–1416 (2020).
- Kather, J. N. et al. Pan-cancer image-based detection of clinically actionable genetic alterations. *Nat. Cancer* **1**, 789–799 (2020).
- Keller, P., Dawood, M. & Minhas, F. U. A. A. Maximum mean discrepancy kernels for predictive and prognostic modeling of whole slide images. In *2023 IEEE 20th International Symposium on Biomedical Imaging (ISBI)* 1–5 (IEEE, 2023); <https://doi.org/10.1109/ISBI53787.2023.10230578>
- Rawat, R. R. et al. Deep learned tissue ‘fingerprints’ classify breast cancers by ER/PR/Her2 status from H&E images. *Sci. Rep.* **10**, 7275 (2020).
- Clark, K. et al. The Cancer Imaging Archive (TCIA): maintaining and operating a public information repository. *J. Digit. Imaging* **26**, 1045–1057 (2013).
- Jahanifar, M. et al. Domain generalization in computational pathology: survey and guidelines. *ACM Comput. Surv.* **57**, 285:1–285:37 (2025).
- Sanchez-Vega, F. et al. Oncogenic signaling pathways in The Cancer Genome Atlas. *Cell* **173**, 321–337 (2018).
- Babur, Ö. et al. Systematic identification of cancer driving signaling pathways based on mutual exclusivity of genomic alterations. *Genome Biol.* **16**, 45 (2015).
- Ciriello, G., Cerami, E., Sander, C. & Schultz, N. Mutual exclusivity analysis identifies oncogenic network modules. *Genome Res.* **22**, 398–406 (2012).
- Ding, T. et al. A multimodal whole-slide foundation model for pathology. *Nat. Med.* **31**, 3749–3761 (2025).
- Tekle, G. E. et al. Co-occurrence and mutual exclusivity: what cross-cancer mutation patterns can tell us. *Trends Cancer* **7**, 823–836 (2021).
- Curtis, C. et al. The genomic and transcriptomic architecture of 2,000 breast tumours reveals novel subgroups. *Nature* **486**, 346–352 (2012).
- Pereira, B. et al. The somatic mutation profiles of 2,433 breast cancers refines their genomic and transcriptomic landscapes. *Nat. Commun.* **7**, 11479 (2016).
- Caso, R. et al. The underlying tumor genomics of predominant histologic subtypes in lung adenocarcinoma. *J. Thorac. Oncol.* **15**, 1844–1856 (2020).
- Weigelt, B. et al. Molecular characterization of endometrial carcinomas in black and white patients reveals disparate drivers with therapeutic implications. *Cancer Discov.* **13**, 2356–2369 (2023).
- Giannakis, M. et al. Genomic correlates of immune-cell infiltrates in colorectal carcinoma. *Cell Rep.* **15**, 857–865 (2016).
- Canisius, S., Martens, J. W. M. & Wessels, L. F. A. A novel independence test for somatic alterations in cancer shows that biology drives mutual exclusivity but chance explains most co-occurrence. *Genome Biol.* **17**, 261 (2016).
- Cerami, E. et al. The cBio cancer genomics portal: an open platform for exploring multidimensional cancer genomics data. *Cancer Discov.* **2**, 401–404 (2012).
- Gao, J. et al. Integrative analysis of complex cancer genomics and clinical profiles using the cBioPortal. *Sci. Signal.* **6**, pl1 (2013).
- Lu, M. Y. et al. Data-efficient and weakly supervised computational pathology on whole-slide images. *Nat. Biomed. Eng.* **5**, 555–570 (2021).
- Dawood, M. et al. Cross-linking breast tumor transcriptomic states and tissue histology. *Cell Rep. Med.* **4**, 101313 (2023).
- Wang, X. et al. Transformer-based unsupervised contrastive learning for histopathological image classification. *Med. Image Anal.* **81**, 102559 (2022).
- Zhang, X., Zhou, X., Lin, M. & Sun, J. ShuffleNet: an extremely efficient convolutional neural network for mobile devices. In *2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition* 6848–6856 (IEEE, 2018); <https://doi.org/10.1109/CVPR.2018.00716>.

36. Russakovsky, O. et al. ImageNet large scale visual recognition challenge. *Int. J. Comput. Vis.* **115**, 211–252 (2015).
37. Mertins, P. et al. Proteogenomics connects somatic mutations to signaling in breast cancer. *Nature* **534**, 55–62 (2016).
38. Carpenter, J. E. & Clarke, C. L. Biobanking sustainability—experiences of the Australian Breast Cancer Tissue Bank (ABCTB). *Biopreserv. Biobank.* **12**, 395–401 (2014).
39. Yule, G. U. Notes on the theory of association of attributes in statistics. *Biometrika* **2**, 121–134 (1903).
40. Bonovas, S. & Piovani, D. Simpson's paradox in clinical research: a cautionary tale. *J. Clin. Med.* **12**, 1633 (2023).
41. Pan, J.-W. et al. The molecular landscape of Asian breast cancers reveals clinically relevant population-specific differences. *Nat. Commun.* **11**, 6433 (2020).
42. Morris, V. K. et al. Phase 1/2 trial of encorafenib, cetuximab, and nivolumab in microsatellite stable BRAFV600E metastatic colorectal cancer. *Cancer Cell* **43**, 2106–2118.e3 (2025).
43. Graham, S. et al. Screening of normal endoscopic large bowel biopsies with interpretable graph learning: a retrospective study. *Gut* **72**, 1709–1721 (2023).
44. Howard, F. M. et al. The impact of site-specific digital histology signatures on deep learning model accuracy and bias. *Nat. Commun.* **12**, 4423 (2021).
45. Schölkopf, B. in *Probabilistic and Causal Inference: The Works of Judea Pearl* 765–804 (Association for Computing Machinery, 2022).
46. Koboldt, D. C. et al. Comprehensive molecular portraits of human breast tumours. *Nature* **490**, 61–70 (2012).
47. Hoadley, K. A. et al. Cell-of-origin patterns dominate the molecular classification of 10,000 tumors from 33 types of cancer. *Cell* **173**, 291–304 (2018).
48. Pocock, J. et al. TIAToolbox as an end-to-end library for advanced tissue image analytics. *Commun. Med.* **2**, 1–14 (2022).
49. Kingma, D. P. & Ba, J. Adam: A method for stochastic optimization. In *International Conference on Learning Representations (ICLR, 2015)*.
50. Ojala, M. & Garriga, G. C. Permutation tests for studying classifier performance. In *2009 Ninth IEEE International Conference on Data Mining* 908–913 (IEEE, 2009); <https://doi.org/10.1109/ICDM.2009.108>
51. Chaibub Neto, E. et al. A permutation approach to assess confounding in machine learning applications for digital health. In *Proc. 25th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* 54–64 (Association for Computing Machinery, 2019); <https://doi.org/10.1145/3292500.3330903>

Acknowledgements

M.D. acknowledges support from the GSK-Warwick PhD Studentship and the Department of Computer Science, University of Warwick. F.M. and N.R. were partially supported by the PathLAKE consortium, which was funded by the Data to Early Diagnosis and Precision Medicine strand of the government's Industrial Strategy Challenge Fund, managed and delivered by UK Research and Innovation (UKRI).

F.M. also acknowledges funding support from EPSRC grant no. EP/W02909X/1. The funders had no role in study design, data collection and analysis, decision to publish or preparation of the manuscript.

Author contributions

F.M. and M.D. designed the study with support from co-authors. F.M. and M.D. wrote the code and analysed the experimental results. N.R., K.B. and F.M. secured the funding. F.M. and N.R. supervised the study. S.T. provided pathological and oncological input. M.D. and F.M. visualized and verified the underlying data. M.D. and F.M. drafted the manuscript with input from co-authors. All authors had full access to all the data in the study and had the final decision to submit for publication.

Competing interests

M.D. conducted this study during his PhD at the University of Warwick, UK. M.D. received PhD studentship support from GSK. K.B. is an employee of GSK. N.R. is the founding Director, CEO and CSO of Histofy Ltd. FM holds shares in Histofy Ltd with no operational involvement. The other authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41551-026-01616-8>.

Correspondence and requests for materials should be addressed to Muhammad Dawood or Fayyaz ul Amir Afsar Minhas.

Peer review information *Nature Biomedical Engineering* thanks Lee Cooper, Nikos Paragios and the other, anonymous, reviewer(s) for their contribution to the peer review of this work. Peer reviewer reports are available.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

© The Author(s) 2026

Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our [Editorial Policies](#) and the [Editorial Policy Checklist](#).

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a Confirmed

- ☐ ☒ The exact sample size (n) for each experimental group/condition, given as a discrete number and unit of measurement
- ☐ ☒ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
- ☐ ☒ The statistical test(s) used AND whether they are one- or two-sided
Only common tests should be described solely by name; describe more complex techniques in the Methods section.
- ☐ ☒ A description of all covariates tested
- ☐ ☒ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
- ☐ ☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
- ☐ ☒ For null hypothesis testing, the test statistic (e.g. F , t , r) with confidence intervals, effect sizes, degrees of freedom and P value noted
Give P values as exact values whenever suitable.
- ☒ ☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
- ☐ ☒ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
- ☐ ☒ Estimates of effect sizes (e.g. Cohen's d , Pearson's r), indicating how they were calculated

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

Data collection

- Whole slide images (WSIs) of TCGA patients used in the study can be downloaded from the NIH Genomic Data Common Portal at this link: <https://portal.gdc.cancer.gov/>.
- The genomic data and clinical data of patients in TCGA, METABRIC, COAD-DFCI, MSK-LUAD and CPTAC cohorts can be downloaded from cBioPortal <https://www.cbioportal.org/>.
- ABCTB data and images were obtained from the Australian Breast Cancer Tissue Bank. The corresponding author could be contacted to facilitate access to ABCTB data.

Data analysis

tiatoolbox (1.3.0) - <https://github.com/TissuelImageAnalytics/tiatoolbox/>
 torch geometric (2.1.0.post1) - <https://pytorch-geometric.readthedocs.io/en/latest/install/installation.html>
 PyTorch (1.12.1+cu102) - <https://pytorch.org/> PRID: SCR_018536
 Scipy (1.7.3) - <http://www.scipy.org/> PRID: SCR_008058
 NumPy (1.22.4) - <http://www.numpy.org/> PRID: SCR_008633
 Pandas (1.5.3) - <https://pandas.pydata.org/> PRID: SCR_018214
 Matplotlib (3.3.4) - <https://matplotlib.org/> PRID: SCR_008624
 OpenCV (4.5.5) - <https://opencv.org/>
 statsmodels (0.12.2) - <https://www.statsmodels.org/> PRID: SCR_016074

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio [guidelines for submitting code & software](#) for further information.

Data

Policy information about [availability of data](#)

All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our [policy](#)

All data used in this study—including histological and molecular information for TCGA and CPTAC cohorts are primarily obtained from the cBioPortal. Detailed manifest files containing unique TCGA or CPTAC identifiers for each sample are provided in our Project GitHub repository: <https://github.com/engrodawood/HistBiases>. Additional information on data access and usage can be found in the repository documentation.

Research involving human participants, their data, or biological material

Policy information about studies with [human participants or human data](#). See also policy information about [sex, gender \(identity/presentation\), and sexual orientation](#) and [race, ethnicity and racism](#).

Reporting on sex and gender	Not applicable
Reporting on race, ethnicity, or other socially relevant groupings	Not applicable
Population characteristics	Detailed population characteristics (e.g., genotypic information, age, etc.) for the samples analyzed in the study can be accessed through cBioportal.
Recruitment	We do not recruit any participants. Instead, we retrospectively analyzed data from 8,221 patients with breast, colorectal, endometrial, and lung cancers across four cohorts: The Cancer Genome Atlas (TCGA), the Molecular Taxonomy of Breast Cancer International Consortium (METABRIC), Memorial Sloan Kettering Cancer Centre (MSK), and Dana-Farber Cancer Institute (DFCI).
Ethics oversight	Biomedical and Scientific Research Ethics Committee (BSREC) University of Warwick Approved the study under application ID BSREC 16/21-22.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see [nature.com/documents/nr-reporting-summary-flat.pdf](https://www.nature.com/documents/nr-reporting-summary-flat.pdf)

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	We analyzed samples from existing datasets and demonstrated the validity of our findings across a large sample size of n = 8,221, derived from multiple datasets
Data exclusions	We excluded Whole Slide Images with missing meta data. The exclusion criteria is clearly explained in the manuscript.
Replication	The results reported in the manuscript can be easily reproduced using the project's GitHub repository: https://github.com/imuhdawood/HistBiases . The repository includes source code for codependency analysis, machine learning models, as well as permutation testing and stratification analysis, along with detailed documentation. These resources can be used to explore potential biases in machine learning models. Additionally, predictions from the models on both training and test datasets are provided, enabling downstream bias analysis without the need to rerun the entire pipeline for each model.
Randomization	Randomization of sample for cross-validation folds (4-folds) was performed at random with stratification. The model is trained on 3 folds and the performance is evaluated on the held out fold.
Blinding	Data of all patients were analyzed anonymously. Not additional blinding was done.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☐	☒ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Clinical data

Policy information about [clinical studies](#)

All manuscripts should comply with the ICMJE [guidelines for publication of clinical research](#) and a completed [CONSORT checklist](#) must be included with all submissions.

Clinical trial registration	Not applicable
Study protocol	Not applicable
Data collection	Not applicable
Outcomes	Not applicable

Plants

Seed stocks	Not applicable
Novel plant genotypes	Not applicable
Authentication	Not applicable